

PHYS 143 – Mechanics and Thermal Physics

Impulse, Momentum and Collisions

*How do forces acting over a time change motion,
and what stays the same when objects interact?*

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 8, §8.1–8.5

By the end of this week, you should be able to:

1. Calculate momentum and impulse, and relate them with the impulse–momentum theorem.
2. Apply conservation of momentum to isolated systems.
3. Solve one- and two-dimensional collision problems, and classify collisions as elastic or inelastic.
4. Use the coefficient of restitution and combine collisions with energy methods.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 DEFINE

What are momentum and impulse?

2 CONSERVE

When is momentum conserved, and what does it say about collisions?

3 EXTEND

What about two dimensions, elasticity and systems?

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. A 0.50 kg ball moves at 4.0 m/s east and then at 2.0 m/s west. Find the change in its velocity.
2. A constant 12 N force acts for 3.0 s on a 4.0 kg object initially at rest. Find its final speed.
3. State Newton's third law. On which bodies do the two forces act?
4. Find the area under a force graph that rises linearly from 0 to 20 N over 0.10 s.
5. Find the kinetic energy of a 2.0 kg object moving at 3.0 m/s.
6. Resolve a 10 m/s velocity at 37° above the $+x$ axis into components.

1. $\Delta v = -2.0 - 4.0 = -6.0 \text{ m/s}$ (taking east as +).
2. $a = 12/4.0 = 3.0 \text{ m/s}^2$; $v = at = 9.0 \text{ m/s}$.
3. Equal and opposite; they act on *different* bodies.
4. $\frac{1}{2}(0.10)(20) = 1.0 \text{ N s}$.
5. $K = \frac{1}{2}(2.0)(3.0)^2 = 9.0 \text{ J}$.
6. $v_x = 10 \cos 37^\circ = 8.0$,
 $v_y = 10 \sin 37^\circ = 6.0 \text{ m/s}$.

Week 3 (Newton's laws), Week 4 (work and kinetic energy) and Lecture 5 (potential energy) are all reused.

Contact 1: Define

What are momentum and impulse?

Week 4

force over a *distance*: work, ΔK .

This week

force over a *time*: impulse, $\Delta \vec{p}$.

Opening question

A 0.15 kg ball moves at 20 m/s and a 0.60 kg ball moves at 5.0 m/s. A catcher stops each with the same 60 N force. Which takes longer to stop? Which stops in the shorter distance?

Predict first

Commit to an answer. Compare mass $\times$ speed for the two balls, and then compare $\frac{1}{2}mv^2$.

The opening question: answer

Momentum mv : ball 1: $0.15 \times 20 = 3.0 \text{ kg m/s}$; ball 2: $0.60 \times 5.0 = 3.0 \text{ kg m/s}$. Equal.

Kinetic energy: ball 1: $\frac{1}{2}(0.15)(20)^2 = 30 \text{ J}$; ball 2: $\frac{1}{2}(0.60)(5.0)^2 = 7.5 \text{ J}$. Different.

Time to stop (force over time)

Same momentum, same force: both need the same time,

$$t = p/F = 3.0/60 = 0.050 \text{ s}.$$

Distance to stop (force over distance)

Distance = K/F : 0.50 m for ball 1, 0.125 m for ball 2.

Momentum governs the *time*; kinetic energy governs the *distance*.

Rule: momentum

$$\vec{p} = m\vec{v} \quad \text{unit: kg m/s}$$

Momentum is a **vector**, in the direction of the velocity. Newton's second law:

$$\sum \vec{F} = \frac{d\vec{p}}{dt}, \text{ which for constant mass is } m\vec{a}.$$

- Components: $p_x = mv_x$, $p_y = mv_y$.
Signs matter.
- The net force is the *rate of change* of momentum.

Common error

Momentum is not kinetic energy. p is a vector and can be negative; K is a scalar and never is.

Rule: impulse–momentum theorem

$$\vec{J} = \int_{t_1}^{t_2} \sum \vec{F} dt = \vec{p}_2 - \vec{p}_1 = \Delta \vec{p} \quad \vec{J} = \vec{F}_{\text{av}} \Delta t$$

The impulse of the net force equals the change in momentum. Unit:
 $\text{Ns} = \text{kg m/s}$.

- Impulse is a **vector**, in the direction of $\Delta \vec{p}$, not a force.
- For a constant force, $\vec{J} = \vec{F} \Delta t$. For a varying force, J is the **area under the F – t graph**.

Two theorems, two questions

	Work–energy theorem	Impulse–momentum theorem
Force acts over	a distance	a time
Quantity delivered	work W (scalar)	impulse $\vec{J}$ (vector)
Changes	kinetic energy K	momentum $\vec{p}$
Use it to find	speed or distance	velocity or time

Predict first

A boxer's glove stops a punch. Which theorem tells you about the force if you know the contact time?

✂ TOOLKIT 1 – Momentum and impulse

1. **AXES:** choose a positive direction before assigning signs (Week 2 Toolkit 1).
2. **STATES:** write $\vec{p}_1 = m\vec{v}_1$ and $\vec{p}_2 = m\vec{v}_2$ with signs, in components if in 2D.
3. **IMPULSE:** $\vec{J} = \Delta\vec{p} = \vec{p}_2 - \vec{p}_1$ (final minus initial).
4. **AVERAGE FORCE:** $F_{av} = J/\Delta t$.

RULE: reversing the velocity changes p by $2mv$, not zero.

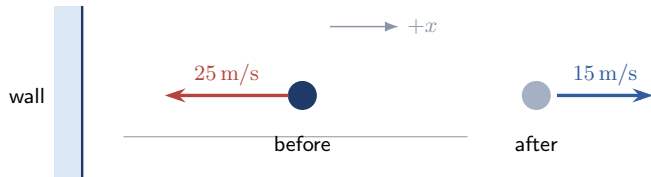
WATCH OUT: a bounce gives a larger impulse than a stop; the impulse is the net force's, so include weight if it matters.

CHECK: does the direction of $\vec{J}$ agree with the direction of the force that must have acted?

Lecturer example: a ball hits a wall

Question

A 0.45 kg ball hits a wall moving horizontally at 25 m/s and rebounds at 15 m/s in the opposite direction. (a) Find the impulse of the net force on the ball. (b) The contact lasts 8.0 ms: find the average force.



Predict first

Which way does the impulse point? Is the average force larger or smaller than the ball's weight (4.4 N)?

A ball hits a wall: solution

Set up: $+x$ away from the wall: $v_{1x} = -25 \text{ m/s}$, $v_{2x} = +15 \text{ m/s}$.

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(a) $J_x = p_{2x} - p_{1x} = m(v_{2x} - v_{1x}) = 0.45 [15 - (-25)] = +18 \text{ N s}$, directed away from the wall.

A ball hits a wall: solution

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(b) $F_{\text{av}} = \frac{J}{\Delta t} = \frac{18}{0.0080} = 2.3 \times 10^3 \text{ N}$.

A ball hits a wall: solution

Set up: $+x$ away from the wall: $v_{1x} = -25 \text{ m/s}$, $v_{2x} = +15 \text{ m/s}$.

(a) $J_x = p_{2x} - p_{1x} = m(v_{2x} - v_{1x}) = 0.45 [15 - (-25)] = +18 \text{ N s}$, directed away from the wall.

(b) $F_{\text{av}} = \frac{J}{\Delta t} = \frac{18}{0.0080} = 2.3 \times 10^3 \text{ N}$.

Evaluate: the impulse points away from the wall, the direction of the force the wall exerts. The force is about 500 times the ball's weight, so ignoring gravity during the contact is justified. ✓

Concept check: why do airbags and crumple zones work?

Question

A 70 kg driver moving at 15 m/s is stopped in a collision. Find the average force if the stopping time is (a) 0.10 s against a rigid dashboard, (b) 0.30 s with an airbag.

Predict first

The impulse is the same in both cases. What changes the force?

Concept check: why do airbags and crumple zones work?

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Predict first

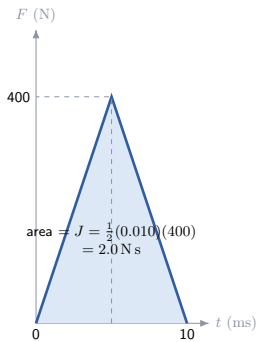
The impulse is the same in both cases. What changes the force?

$J = \Delta p = 70(0 - 15) = -1050 \text{ N s}$ in both cases.

(a) $F_{\text{av}} = 1050/0.10 = 1.1 \times 10^4 \text{ N}$. (b) $F_{\text{av}} = 1050/0.30 = 3.5 \times 10^3 \text{ N}$.

The same impulse delivered over three times the time gives one third of the force: that is why airbags, crumple zones and bending your knees on landing work.

Impulse from a force–time graph



A tennis serve: the racket exerts the force shown on a 0.057 kg ball, initially at rest.

$$J = \text{area} = \frac{1}{2}(0.010)(400) = 2.0 \text{ N s.}$$

$$v = J/m = 2.0/0.057 = 35 \text{ m/s.}$$

$$F_{\text{av}} = J/\Delta t = 2.0/0.010 = 200 \text{ N, half the peak.}$$



TOOLKIT 2 – Impulse–momentum theorem

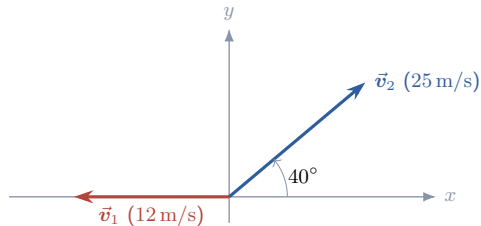
1. Sketch F against t or find Δp . 2. Impulse = area (or Δp). 3. $F_{\text{av}} = J/\Delta t$. 4. Use $J = \Delta p$ to find v or Δt .

WATCH OUT: average force is not the peak force. **CHECK:** do the units work: $\text{N s} = \text{kg m/s}$?

Lecturer example: a kicked ball in two dimensions

Question

A 0.45 kg ball moving left at 12 m/s is kicked and leaves at 25 m/s, 40° above the horizontal and to the right. The contact lasts 12 ms. Find the impulse and the average force.



Predict first

Which direction is $\vec{J}$: mainly right, mainly up, or between?

Set up: $+x$ right, $+y$ up. $v_{1x} = -12$, $v_{1y} = 0$; $v_{2x} = 25 \cos 40^\circ = 19.2$,
 $v_{2y} = 25 \sin 40^\circ = 16.1 \text{ m/s}$.

Set up: $+x$ right, $+y$ up. $v_{1x} = -12$, $v_{1y} = 0$; $v_{2x} = 25 \cos 40^\circ = 19.2$,
 $v_{2y} = 25 \sin 40^\circ = 16.1 \text{ m/s}$.

Execute (components, Week 1):

$$J_x = m(v_{2x} - v_{1x}) = 0.45(19.2 + 12) = 14.0 \text{ N s} \quad J_y = m(v_{2y} - v_{1y}) = 0.45(16.1) = 7.2 \text{ N s}$$
$$J = \sqrt{14.0^2 + 7.2^2} = 15.8 \text{ N s at } \tan^{-1}(7.2/14.0) = 27^\circ \text{ above the horizontal (quadrant I).}$$
$$F_{\text{av}} = J/\Delta t = 15.8/0.012 = 1.3 \times 10^3 \text{ N in the same direction.}$$

Set up: $+x$ right, $+y$ up. $v_{1x} = -12$, $v_{1y} = 0$; $v_{2x} = 25 \cos 40^\circ = 19.2$,
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$$F_{\text{av}} = J/\Delta t = 15.8/0.012 = 1.3 \times 10^3 \text{ N in the same direction.}$$

Evaluate: between right and up, mainly right, as predicted. $\vec{J}$ is the *vector* difference $\vec{p}_2 - \vec{p}_1$, not the difference of magnitudes. ✓

Your turn: a tennis ball and a racket

Your turn

A 0.058 kg tennis ball arrives horizontally at 30 m/s and leaves the racket in the opposite direction at 45 m/s. The contact lasts 5.0 ms. Find the impulse and the average force on the ball.

Predict first

Sketch with $+x$ along the outgoing velocity. Sign of J ? Roughly how many times the ball's weight?

Your turn

A 0.058 kg tennis ball arrives horizontally at 30 m/s and leaves the racket in the opposite direction at 45 m/s. The contact lasts 5.0 ms. Find the impulse and the average force on the ball.

Predict first

Sketch with $+x$ along the outgoing velocity. Sign of J ? Roughly how many times the ball's weight?

$$J = m(v_2 - v_1) = 0.058 [45 - (-30)] = 4.4 \text{ N s along the outgoing direction.}$$

$$F_{\text{av}} = 4.4/0.0050 = 8.7 \times 10^2 \text{ N.}$$

Evaluate: positive in the chosen direction; about 1500 times the weight (0.57 N). ✓

Contact 1: what we can now do

Force over time

↓

Impulse $\vec{J} = \int \sum \vec{F} dt$

↓

$\vec{J} = \Delta \vec{p}$

You should be able to

- calculate momentum and impulse as vectors;
- find impulse from Δp or a force graph;
- find an average collision force;
- distinguish momentum from kinetic energy.

✓ Before Contact 2

What happens to the total momentum of two bodies that push on each other?
Read Young and Freedman, §8.2.

Contact 2: Conserve

*When is momentum conserved, and
what does it say about collisions?*

Poll

Two astronauts at rest in space touch and push each other apart. A has mass 60 kg and B has mass 90 kg. What is the total momentum afterwards? Which astronaut moves faster?

Predict first

Which law connects the push on A to the push on B ? What does it say about $\Delta\vec{p}_A$ and $\Delta\vec{p}_B$?

Why momentum is conserved

Two bodies interacting only with each other. By Newton's third law $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$, for the same time Δt :

$$\Delta \vec{p}_B = \vec{F}_{A \text{ on } B} \Delta t, \quad \Delta \vec{p}_A = \vec{F}_{B \text{ on } A} \Delta t = -\Delta \vec{p}_B$$

so $\Delta(\vec{p}_A + \vec{p}_B) = 0$.

Rule: conservation of momentum

If the net *external* force on a system is zero (an **isolated** system), its total momentum is constant:

$$\vec{P} = \vec{p}_A + \vec{p}_B + \cdots = \text{constant}, \quad \vec{P}_{\text{before}} = \vec{P}_{\text{after}}$$

TOOLKIT 3 – Conservation of momentum

1. **SYSTEM:** choose the objects so that the important forces are *internal*.
2. **ISOLATED?** Check the external forces. During a short collision they are usually negligible compared with the collision forces (the *impulse approximation*).
3. **SKETCH** before and after, with $+x$, $+y$ and each velocity labelled.
4. **WRITE** one equation per axis: $\sum p_{x,\text{before}} = \sum p_{x,\text{after}}$, and the same for y .
5. **SOLVE** for the unknown velocity; check the signs.

WATCH OUT: momentum is a vector; ignoring signs or components is the most common error.

CHECK: do the units match, and does the direction make physical sense?

Question

A 3.0 kg rifle, initially at rest, fires a 0.010 kg bullet at 400 m/s. Find the rifle's recoil velocity, and compare the kinetic energies of rifle and bullet.

Predict first

Which moves faster, bullet or rifle? Which has more kinetic energy?

Question

A 3.0 kg rifle, initially at rest, fires a 0.010 kg bullet at 400 m/s. Find the rifle's recoil velocity, and compare the kinetic energies of rifle and bullet.

Predict first

Which moves faster, bullet or rifle? Which has more kinetic energy?

Set up: system: rifle and bullet; external horizontal force negligible. $+x$ along the bullet's velocity; $P_{\text{before}} = 0$.

Execute: $0 = m_b v_b + m_r v_r \Rightarrow v_r = -\frac{(0.010)(400)}{3.0} = -1.3 \text{ m/s}$.

$K_b = \frac{1}{2}(0.010)(400)^2 = 800 \text{ J}$; $K_r = \frac{1}{2}(3.0)(1.33)^2 = 2.7 \text{ J}$.

Evaluate: the momenta are equal and opposite, but the light bullet carries almost all the kinetic energy: $K = p^2/(2m)$. ✓

Rule: collisions of an isolated system

Momentum is conserved in **every** collision. **Kinetic energy** is conserved only in an **elastic** collision.

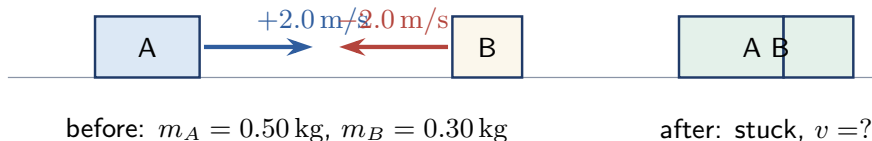
Type	Momentum	Kinetic energy
elastic	conserved	$K_{\text{after}} = K_{\text{before}}$
inelastic	conserved	$K_{\text{after}} < K_{\text{before}}$
completely inelastic	conserved	maximum loss; objects stick together

Common error

“Momentum is lost in an inelastic collision.” It is not: the kinetic energy is partly converted into other forms (deformation, heat, sound).

Lecturer example: gliders that stick together

Two gliders on a frictionless air track collide head on and stick together. Find the final velocity and the kinetic energy lost.



Predict first

Which direction will they move? Will the final speed be larger or smaller than 2.0 m/s ?

Gliders that stick together: solution

Set up: $+x$ to the right. $m_A v_A + m_B v_B = (m_A + m_B)v$.

Gliders that stick together: solution

Set up: $+x$ to the right. $m_A v_A + m_B v_B = (m_A + m_B)v$.

Execute: $(0.50)(2.0) + (0.30)(-2.0) = (0.80)v \Rightarrow v = \frac{1.0 - 0.60}{0.80} = +0.50 \text{ m/s}$ (to the right).

$$K_{\text{before}} = \frac{1}{2}(0.50)(2.0)^2 + \frac{1}{2}(0.30)(2.0)^2 = 1.6 \text{ J}; \quad K_{\text{after}} = \frac{1}{2}(0.80)(0.50)^2 = 0.10 \text{ J}.$$

Lost: 1.5 J, about 94%.

Gliders that stick together: solution

Set up: $+x$ to the right. $m_A v_A + m_B v_B = (m_A + m_B)v$.

Execute: $(0.50)(2.0) + (0.30)(-2.0) = (0.80)v \Rightarrow v = \frac{1.0 - 0.60}{0.80} = +0.50 \text{ m/s}$ (to the right).

$$K_{\text{before}} = \frac{1}{2}(0.50)(2.0)^2 + \frac{1}{2}(0.30)(2.0)^2 = 1.6 \text{ J}; \quad K_{\text{after}} = \frac{1}{2}(0.80)(0.50)^2 = 0.10 \text{ J}.$$

Lost: 1.5 J, about 94%.

Evaluate: right (the heavier glider's way), slower than 2.0 m/s, as predicted; $K_{\text{after}} < K_{\text{before}}$, so the collision is inelastic. ✓

TOOLKIT 4 – Collisions in one dimension

1. **SKETCH** before and after, with the $+x$ axis and signed velocities.
 2. **MOMENTUM:** always write $m_1v_1 + m_2v_2 = m_1v'_1 + m_2v'_2$.
 3. **TYPE:** stuck together $\Rightarrow$ one common v' ; elastic $\Rightarrow$ add $K_{\text{before}} = K_{\text{after}}$; otherwise give e or one final velocity.
 4. **SOLVE** the simultaneous equations (Week 3 refresher).
 5. **CHECK K :** compute K before and after to classify the collision.
- WATCH OUT:** $K_{\text{after}} > K_{\text{before}}$ is impossible for an isolated collision: look for an arithmetic or sign error.
- CHECK:** is momentum conserved when you substitute your answers?

Your turn

A 0.40 kg glider moving at $+3.0 \text{ m/s}$ collides head on with a 0.60 kg glider moving at -1.0 m/s . After the collision the 0.60 kg glider moves at $+1.5 \text{ m/s}$. Find the velocity of the 0.40 kg glider, and classify the collision.

Predict first

Will the 0.40 kg glider rebound, or continue forward? Is K conserved?

Your turn

A 0.40 kg glider moving at +3.0 m/s collides head on with a 0.60 kg glider moving at -1.0 m/s. After the collision the 0.60 kg glider moves at +1.5 m/s. Find the velocity of the 0.40 kg glider, and classify the collision.

Predict first

Will the 0.40 kg glider rebound, or continue forward? Is K conserved?

$$(0.40)(3.0) + (0.60)(-1.0) = 0.40 v'_A + (0.60)(1.5) \Rightarrow 0.60 = 0.40 v'_A + 0.90 \Rightarrow v'_A = -0.75 \text{ m/s (it rebounds).}$$

$$K_{\text{before}} = 2.1 \text{ J}, \quad K_{\text{after}} = 0.11 + 0.68 = 0.79 \text{ J: } \mathbf{inelastic} \text{ (but not completely).}$$

Evaluate: K decreased and momentum is conserved. ✓

Rule: target at rest, elastic

$$v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1 \quad v_2' = \frac{2m_1}{m_1 + m_2} v_1$$

(from momentum and kinetic-energy conservation; derivation in the supplement)



before: masses in kg, target at rest

after

$$m_1 = 1.0 \text{ kg at } 4.0 \text{ m/s hits } m_2 = 3.0 \text{ kg at rest: } v_1' = \frac{1.0 - 3.0}{4.0} (4.0) = -2.0 \text{ m/s,}$$

$$v_2' = \frac{2(1.0)}{4.0} (4.0) = +2.0 \text{ m/s.}$$

$$\text{Check: } K \text{ before} = 8.0 \text{ J; after} = \frac{1}{2}(1.0)(2.0)^2 + \frac{1}{2}(3.0)(2.0)^2 = 8.0 \text{ J. } \checkmark$$

Special cases of an elastic collision

Equal masses, target at rest

$v'_1 = 0$, $v'_2 = v_1$: the moving object stops and the target moves off with the same velocity.

Very heavy target

$m_2 \gg m_1$: $v'_1 \approx -v_1$, $v'_2 \approx 0$: the light object bounces back with the same speed.

Very light target

$m_2 \ll m_1$: $v'_1 \approx v_1$, $v'_2 \approx 2v_1$: the heavy object barely slows; the light one shoots off at about twice the speed.

Predict first

A ping-pong ball hits a stationary bowling ball elastically. What happens to the ping-pong ball?

Isolated system



$$\vec{P}_{\text{before}} = \vec{P}_{\text{after}}$$



Collision types (check K)

You should be able to

- decide whether momentum is conserved;
- solve recoil and 1D collisions;
- classify a collision from its kinetic energy;
- solve elastic collisions in 1D.

✓ Before Contact 3

What if the objects move off at angles? Read Young and Freedman, §8.3–8.5.
Recall potential energy from Lecture 5, $U = mgy$.

Contact 3: Extend

What about two dimensions, elasticity and systems?

Rule: components are independent

Momentum is conserved **separately along each axis**:

$$\sum p_{x,\text{before}} = \sum p_{x,\text{after}} \quad \sum p_{y,\text{before}} = \sum p_{y,\text{after}}$$

Predict first

Puck *A* moving along $+x$ hits an identical puck *B* at rest and moves off at 30° above the x -axis. In which direction must *B* move? Above or below the axis?

Rule: components are independent

Momentum is conserved **separately along each axis**:

$$\sum p_{x,\text{before}} = \sum p_{x,\text{after}} \quad \sum p_{y,\text{before}} = \sum p_{y,\text{after}}$$

Predict first

Puck *A* moving along $+x$ hits an identical puck *B* at rest and moves off at 30° above the x -axis. In which direction must *B* move? Above or below the axis?

The total p_y is zero before the collision, so p_y after must also be zero: *B* moves **below** the axis.

TOOLKIT 5 – Collisions in two dimensions

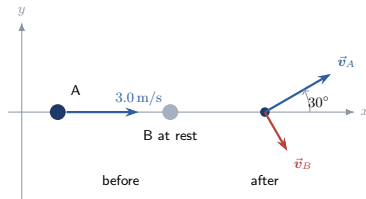
1. **SKETCH** before and after, with axes and angles from $+x$ (Week 1).
2. **RESOLVE** each velocity into components; assign signs.
3. **WRITE** two equations: $\sum p_x$ conserved, $\sum p_y$ conserved.
4. **SOLVE** for the two unknown components; then reconstruct the speed and direction (Week 1 Toolkit 4).
5. **CHECK** K to classify the collision.

WATCH OUT: y momentum before need not be zero; include every component, with signs.

CHECK: do the final components give a direction that makes sense with the sketch?

Lecturer example: a glancing collision

A 0.20 kg puck A moving at 3.0 m/s along $+x$ hits an identical puck B at rest. A leaves at 2.6 m/s, 30° above the x -axis. Find B 's velocity and classify the collision.



Predict first

Sign of v_{By} ? Is the speed of B more or less than 2.6 m/s?

A glancing collision: solution

Set up: equal masses cancel. x : $3.0 = 2.6 \cos 30^\circ + v_{Bx}$; y : $0 = 2.6 \sin 30^\circ + v_{By}$.

A glancing collision: solution

Set up: equal masses cancel. x : $3.0 = 2.6 \cos 30^\circ + v_{Bx}$; y : $0 = 2.6 \sin 30^\circ + v_{By}$.

Execute: $v_{Bx} = 3.0 - 2.25 = 0.75 \text{ m/s}$; $v_{By} = -1.3 \text{ m/s}$.

$v_B = \sqrt{0.75^2 + 1.3^2} = 1.5 \text{ m/s}$ at $\tan^{-1}(1.3/0.75) = 60^\circ$ **below** the x -axis.

A glancing collision: solution

Set up: equal masses cancel. x : $3.0 = 2.6 \cos 30^\circ + v_{Bx}$; y : $0 = 2.6 \sin 30^\circ + v_{By}$.

Execute: $v_{Bx} = 3.0 - 2.25 = 0.75 \text{ m/s}$; $v_{By} = -1.3 \text{ m/s}$.

$v_B = \sqrt{0.75^2 + 1.3^2} = 1.5 \text{ m/s}$ at $\tan^{-1}(1.3/0.75) = 60^\circ$ **below** the x -axis.

Classify: $K_{\text{before}} = \frac{1}{2}(0.20)(3.0)^2 = 0.90 \text{ J}$; $K_{\text{after}} = \frac{1}{2}(0.20)(2.6^2 + 1.5^2) = 0.90 \text{ J}$: **elastic**.

Evaluate: $v_{By} < 0$ and slower than A , as predicted; the angle between the paths is 90° , as expected for an equal-mass elastic collision. ✓

The coefficient of restitution

Rule: coefficient of restitution

$$e = \frac{\text{relative speed of separation}}{\text{relative speed of approach}} = \frac{v'_2 - v'_1}{v_1 - v_2} \quad (0 \leq e \leq 1)$$

$e = 1$: elastic. $e = 0$: completely inelastic (objects stick). $0 < e < 1$: inelastic.

In an elastic collision the relative velocity simply **reverses**: $v_1 - v_2 = -(v'_1 - v'_2)$.

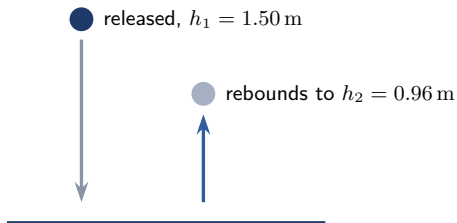
Common error

“Elastic means no deformation.” It means no kinetic energy is lost: the deformation is temporary and fully recovered.

TOOLKIT 6 – Restitution and relative velocity

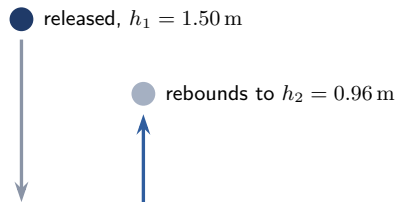
1. **APPROACH AND SEPARATION:** $v_{\text{app}} = v_1 - v_2$ (before), $v_{\text{sep}} = v'_2 - v'_1$ (after).
 2. $e = v_{\text{sep}}/v_{\text{app}}$. For a ball dropped on the floor: $e = \sqrt{h_2/h_1}$.
 3. **COMBINE** the e equation with momentum: two equations for two unknown velocities.
- WATCH OUT:** use signed velocities in $v_1 - v_2$; the separation is positive when the objects move apart.
- CHECK:** is $0 \leq e \leq 1$ for your result, and is $K_{\text{after}} \leq K_{\text{before}}$?

Lecturer example: a bouncing ball and a glider



- (a) A ball is dropped from 1.50 m and rebounds to 0.96 m . Find e .
- (b) A 2.0 kg glider at 3.0 m/s hits a 1.0 kg glider at rest with $e = 0.50$. Find both velocities after the collision.

Lecturer example: a bouncing ball and a glider



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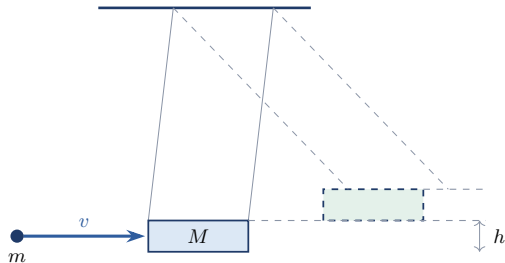
(b) A 2.0 kg glider at 3.0 m/s hits a 1.0 kg glider at rest with $e = 0.50$. Find both velocities after the collision.

(a) $v_1 = \sqrt{2gh_1} = 5.4$ m/s, $v_2 = \sqrt{2gh_2} = 4.3$ m/s, $e = v_2/v_1 = 0.80$ (equivalently $\sqrt{h_2/h_1}$).

(b) Momentum: $6.0 = 2.0v'_1 + 1.0v'_2$. Restitution: $v'_2 - v'_1 = 0.50(3.0) = 1.5$. So $6.0 = 3.0v'_1 + 1.5$: $v'_1 = 1.5$ m/s, $v'_2 = 3.0$ m/s.

K : before 9.0 J, after $4.5 + 2.25 = 6.75$ J: inelastic, with $e < 1$. ✓

Collision, then energy: the ballistic pendulum



Rule: two stages

Stage 1 (the collision): momentum is conserved; kinetic energy is **not**.

Stage 2 (the swing): mechanical energy is conserved (Lecture 5).

Predict first

Why can we not apply energy conservation through the collision?

TOOLKIT 7 – Two-stage problems: collision, then energy

1. **SPLIT** the motion into stages at the instant the collision ends.
2. **STAGE 1 (collision)**: momentum conservation, with the objects stuck or the velocities given. Do not use K conservation unless the collision is elastic.
3. **STAGE 2 (motion after)**: work–energy theorem or mechanical energy conservation (with friction or a spring if present).
4. **LINK** through the velocity just after the collision.

WATCH OUT: using one equation (energy or momentum) across both stages.

CHECK: is the kinetic energy lost in stage 1 consistent with an inelastic collision?

Question

A 10 g bullet strikes a 2.0 kg block hanging on strings and embeds in it. The block and bullet swing up through $h = 0.20$ m. Find the speed of the bullet, and the fraction of its kinetic energy lost in the collision.

Predict first

Roughly: is the bullet speed of order 10^1 , 10^2 or 10^3 m/s? Is most of the kinetic energy lost?

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Predict first

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Stage 2: $\frac{1}{2}(M + m)V^2 = (M + m)gh \Rightarrow V = \sqrt{2gh} = \sqrt{2(9.8)(0.20)} = 1.98$ m/s.

Stage 1: $mv = (M + m)V \Rightarrow v = \frac{2.01(1.98)}{0.010} = 4.0 \times 10^2$ m/s.

Energy: $K_{\text{before}} = \frac{1}{2}(0.010)(398)^2 = 790$ J; $K_{\text{after}} = \frac{1}{2}(2.01)(1.98)^2 = 3.9$ J, so 99.5% is lost.

✓

Rule: centre of mass

$$x_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2 + \cdots}{m_1 + m_2 + \cdots}$$

$$\vec{P} = M\vec{v}_{\text{cm}} \quad \sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}}$$

The centre of mass moves as if all the mass were at that point and all external forces acted there. Internal forces cannot move it. The full treatment follows in the lecture on rotation.

Physical meaning

If no external horizontal force acts, $\vec{v}_{\text{cm}}$ is constant; if the system starts at rest, the centre of mass **stays where it is**.

Lecturer example: a tug-of-war on ice

Two people, 60 kg and 90 kg, stand 10 m apart on frictionless ice and pull on a rope between them. Where do they meet?



Predict first

Who moves farther? Does the centre of mass move?

Lecturer example: a tug-of-war on ice

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Predict first

Who moves farther? Does the centre of mass move?

$$x_{\text{cm}} = \frac{60(0) + 90(10)}{150} = 6.0 \text{ m from the 60 kg person.}$$
 No external horizontal force, starting at rest: the centre of mass stays put, so they meet there.
The lighter person moves 6.0 m; the heavier moves 4.0 m. ✓

Integrated problem: a bullet in a block on a rough floor

Decide what physics applies; do not look for a formula

A 20 g bullet moving horizontally at 300 m/s embeds in a 1.98 kg block at rest on a rough floor ($\mu_k = 0.25$).

- (a) What is the speed of the block and bullet just after the impact?
- (b) How far do they slide?
- (c) What fraction of the bullet's kinetic energy is lost in the collision?

Predict first

IDENTIFY $\rightarrow$ stages and system $\rightarrow$ PREDICT $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE. Is the sliding distance more or less than 10 m?

Identify, set up: two stages (Toolkit 7): the collision (momentum), then sliding with friction (work–energy).

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(a) Stage 1: $(0.020)(300) = (2.00)V \Rightarrow V = 3.0 \text{ m/s}$.

(b) Stage 2: $-\mu_k M g d = 0 - \frac{1}{2} M V^2 \Rightarrow d = \frac{V^2}{2\mu_k g} = \frac{9.0}{2(0.25)(9.8)} = 1.8 \text{ m}$.

Identify, set up: two stages (Toolkit 7): the collision (momentum), then sliding with friction (work–energy).

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(c) Energy: $K_{\text{before}} = \frac{1}{2}(0.020)(300)^2 = 900 \text{ J}; \quad K_{\text{after}} = \frac{1}{2}(2.00)(3.0)^2 = 9.0 \text{ J}.$ Lost: 99%.

Identify, set up: two stages (Toolkit 7): the collision (momentum), then sliding with friction (work–energy).

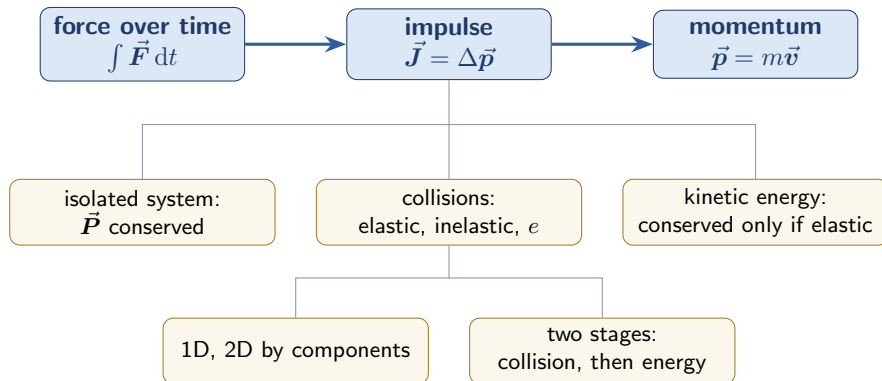
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Evaluate: $d < 10 \text{ m}$, as predicted; the heavy block moves slowly after absorbing the bullet's momentum, but most of the energy goes into deformation and heat in the collision. ✓

Lecture 7 concept map



MASTER PROCEDURE – MOMENTUM

- 1. IDENTIFY** The system, the time interval and what is asked.
 - 2. SKETCH** Before and after (and any later stage), with axes and signed velocities.
 - 3. ISOLATED?** Check the external forces; use the impulse approximation for short collisions.
 - 4. PREDICT** Directions, speeds, and the trend of the kinetic energy.
 - 5. SELECT THE MODEL** Impulse–momentum, conservation, collision type, restitution, or a second stage with energy.
 - 6. EXECUTE** One equation per axis; add e or K as needed.
 - 7. EVALUATE** Units, signs, $K_{\text{after}} \leq K_{\text{before}}$, momentum check.
- SKETCH → PREDICT → CALCULATE → CHECK.

Inside the course framework

IDENTIFY ↔ steps 1–3
SET UP ↔ steps 4–5
EXECUTE ↔ step 6
EVALUATE ↔ step 7

Physical meaning

Momentum conservation always holds for an isolated system; energy methods only when you know what forces do work.

THE SEVEN TOOLKITS

- | | | |
|---|--|---|
| 1 | Momentum and impulse | vectors; $\vec{J} = \Delta\vec{p}$; signs and axes |
| 2 | Impulse–momentum theorem | F – t area; average force = $J/\Delta t$ |
| 3 | Conservation of momentum | system; isolated; before and after; per axis |
| 4 | Collisions in one dimension | types; momentum always; K check |
| 5 | Collisions in two dimensions | components; reconstruct speed and angle |
| 6 | Restitution and relative velocity | $e = v_{\text{sep}}/v_{\text{app}}$; momentum plus e |
| 7 | Two-stage problems | collision (momentum), then energy |

The Master Procedure sits above these. Earlier toolkits (components, free-body diagrams, work–energy, mechanical energy) are reused, not renamed.

Momentum

$$\vec{p} = m\vec{v}, \quad \sum \vec{F} = d\vec{p}/dt$$

Impulse

$$\vec{J} = \int \sum \vec{F} dt = \Delta\vec{p} = \vec{F}_{\text{av}}\Delta t$$

Conservation (isolated)

$$\vec{P}_{\text{before}} = \vec{P}_{\text{after}} \text{ (each axis)}$$

Collisions

elastic: K conserved; inelastic: K decreases

Restitution

$$e = (v'_2 - v'_1)/(v_1 - v_2)$$

Centre of mass

$$x_{\text{cm}} = \sum m_i x_i / M, \quad \vec{P} = M\vec{v}_{\text{cm}}$$

Six habits for Lecture 7

✓ Habit 1

Choose the axes and signs, and sketch before and after.

✓ Habit 2

Treat momentum as a vector: one equation per axis.

✓ Habit 3

Check that the system is isolated before conserving momentum.

✓ Habit 4

Momentum is always conserved in an isolated collision; kinetic energy is not.

✓ Habit 5

Split two-stage problems: collision, then energy.

✓ Habit 6

Compute K before and after, and compare with your prediction.

Checklist: can I ...

- ☐ distinguish momentum from kinetic energy?
- ☐ find impulse from $\Delta\vec{p}$ and from a force graph?
- ☐ compute an average collision force?
- ☐ decide whether momentum is conserved?
- ☐ solve recoil and one-dimensional collisions?
- ☐ classify a collision from its kinetic energy?
- ☐ solve a two-dimensional collision by components?
- ☐ use the coefficient of restitution?
- ☐ solve a ballistic pendulum in two stages?
- ☐ state $\vec{P} = M\vec{v}_{\text{cm}}$ and find a centre of mass?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra impulse, conservation, collision, restitution and two-stage problems. Supplementary notes: the derivation of the elastic-collision formulas, the impulse theorem for a variable force, rocket propulsion and more on the centre of mass.